

7. Write a function that checks whether a string is a palindrome.
8. Create a function to sort a list without using built-in sort functions.

1. Write a function to calculate the factorial of a number.

```
def factorial(n):
    fact=1
    for i in range(1,n+1):
        fact=fact*i
    return fact
n=int(input("Enter a number:"))
print("Factorial:",factorial(n))
```

output:
Enter a number:5
Factorial: 120

2. Write a function to check if a number is prime.

```
def is_prime(n):
    if n <= 1:
        return False

    for i in range(2, n):
        if n % i == 0:
            return False

    return True

n = int(input("Enter a number: "))

if is_prime(n):
    print("Prime Number")
else:
    print("Not a Prime Number")
```

output:
Enter a number: 7
Prime Number

3. Write a function that takes a string and returns the number of vowels.

```
def count_vowels(s):
    count = 0

    for i in s:
        if i.lower() in "aeiou":
            count += 1
```

```
    return count
```

```
s = input("Enter a string: ")  
print("Number of vowels:", count_vowels(s))
```

output:

```
Enter a string: hello world  
Number of vowels: 3
```

4. Write a function that returns the nth Fibonacci number.

```
def fibonacci(n):  
    a = 0  
    b = 1  
  
    for i in range(n):  
        a, b = b, a + b  
  
    return a
```

```
n = int(input("Enter n: "))  
print("Fibonacci number:", fibonacci(n))
```

output:

```
Enter n: 8  
Fibonacci number: 21
```

5. Create a calculator function that takes two numbers and an operator (+, -, *, /) and returns the result.

```
def calculator(a, b, operator):  
    if operator == "+":  
        return a + b  
    elif operator == "-":  
        return a - b  
    elif operator == "*":  
        return a * b  
    elif operator == "/":  
        return a / b  
    else:  
        return "Invalid operator"
```

```
a = float(input("Enter first number: "))  
b = float(input("Enter second number: "))  
operator = input("Enter operator (+, -, *, /): ")
```

```
print("Result:", calculator(a, b, operator))
```

output:

```
Enter first number: 10
Enter second number: 6
Enter operator (+, -, *, /): +
Result: 16.0
```

6. Write a function to count the number of words in a sentence.

```
def count_words(sentence):
    words = sentence.split()
    return len(words)

sentence = input("Enter a sentence: ")
print("Number of words:", count_words(sentence))
```

output:

```
Enter a sentence: python is a programming language
Number of words: 5
```

7. Write a function that checks whether a string is a palindrome.

```
def is_palindrome(s):
    if s == s[::-1]:
        return True
    else:
        return False

s = input("Enter a string: ")

if is_palindrome(s):
    print("Palindrome")
else:
    print("Not a Palindrome")
```

output:

```
Enter a string: madam
Palindrome
```

8. Create a function to sort a list without using built-in sort functions.

```
def sort_list(numbers):
    n = len(numbers)
```

```
for i in range(n):
    for j in range(0, n - i - 1):
        if numbers[j] > numbers[j + 1]:
            numbers[j], numbers[j + 1] = numbers[j + 1], numbers[j]

    return numbers
```

```
numbers = [5, 2, 8, 1, 3]
```

```
print("Original list:", numbers)
print("Sorted list:", sort_list(numbers))
```

output:

```
Original list: [5, 2, 8, 1, 3]
Sorted list: [1, 2, 3, 5, 8]
```